

TSA SERIES

MODEL

TSA700-6N

The TSA700-6N is a horizontal-multi-joint SCARA robot built for repetitive planar work — assembly, part loading, sorting, inspection and dispensing. 6 kg payload, 700 mm arm reach, 200 mm Z stroke and a fast 0.44 s standard cycle, in a compact 24 kg body. Integrated pneumatic and signal harness through the arm keeps end-effector tooling clean and repeatable.

4-Axis SCARA	Repeatability	Texsonics RC	Site-ready
Payload 6 kg · Reach 700 mm	±0.02 mm (XY) positional accuracy	In-house controller, drives and teach pendant	Floor / Wall / Ceiling mount, factory calibrated

01 SPECIFICATIONS

MECHANICAL		ENVIRONMENT / CONTROL	
AXES	4 (SCARA)	SIGNALS	15 pin D-Sub (13-return cable)
PAYLOAD	6 kg	PNEUMATICS	1 × Ø6 + 2 × Ø6 (0.59 MPa)
ARM LENGTH	700 mm	HARNESS LENGTH	3 m
Z STROKE	200 mm	PROTECTION	IP20
STANDARD CYCLE	0.44 s	ENVIRONMENT	0–40 °C, 35–85 % RH
BODY WEIGHT	24 kg		

02 PRODUCT



TSA700-6N · TSA SERIES

03

WORKING RANGE

MOTION — GROUP A		MOTION — GROUP B	
A-AXIS (INNER ARM)	±140°	Z MAX SPEED	1100 mm/s
B-AXIS (OUTER ARM)	±145°	W MAX SPEED	2000°/s
Z-AXIS (VERTICAL STROKE)	200 mm	REPEATABILITY — A/B	±0.02 mm
W-AXIS (TOOL ROTATION)	±360°	REPEATABILITY — Z	±0.01 mm
A/B MAX SPEED	8700 mm/s	REPEATABILITY — W	±0.01°

04

OVERVIEW

HIGH-THROUGHPUT
PLANAR MOTION,
0.44 S CYCLE.

The TSA700-6N is a 4-axis SCARA built for repetitive planar work — assembly, part loading, sorting, inspection and dispensing. Integrated pneumatic and signal harness keeps end-effector tooling clean and repeatable.

01

FAST CYCLE
0.44 s standard cycle, 8700 mm/s XY speed and 2000°/s tool rotation for the shortest possible takt times.

02

HARNESS-THROUGH ARM
15-pin D-Sub signal and dual 6 mm pneumatic lines routed through the arm keep tooling wiring clean.

03

PLANAR ACCURACY
±0.02 mm XY, ±0.01 mm Z, ±0.01° tool — precision suited to assembly and inspection cells.

05

CONTROL & INTEGRATION

CONTROLLER Texsonics RC series in-house motion controller, dual-kernel real-time OS	TEACH PENDANT 11.5" touch tablet drag-teaching, Wi-Fi, on-device offline programming	FIELDBUS EtherCAT / Modbus TCP industrial fieldbus, dual-protocol support	SOFTWARE TEXCAM offline CAM toolpath generation, digital twin simulation, vision
--	--	--	---

06

APPLICATIONS

Deployable across the primary applications the TSA700-6N was engineered for — full production-line integration with the Texsonics control stack.

Assembly	Loading / Unloading	Sorting	Inspection	Dispensing	Machine Tending
CNC Loading	Palletizing	Packaging	Vision Inspection		

07

BUILT FOR THE OPERATOR

- Live 3D position display
- Online command execution
- Position editor
- Variable monitor
- I/O monitor
- Error diagnostics
- Dynamic teach-in
- Digital twin simulation

08

SERVICE & SUPPORT**Made in Coimbatore.
Supported from
Coimbatore.**

Every TSA700-6N is engineered, built and calibrated in our Coimbatore facility. Field service, spares and applications engineering ship from the same address — no import lead times, no time-zone escalations.

STANDARD WARRANTY

12 months on the mechanical assembly, 12 months on the controller.

APPLICATIONS ENGINEERING

Cell design, cycle-time studies and end-of-arm tooling by our in-house team, included on every project.

SPARES & FIELD SERVICE

Direct-to-plant spares from Coimbatore, with pan-India field service engineers on 24-hour dispatch.

09

READY TO INTEGRATE**Talk to an engineer about
integrating the TSA700-6N
into your line.**

Applications engineering, cycle-time sizing, and turnkey cell design are included on every project.

CONTACT**Dharmar R**

Managing Director

+91 94426 24304

dharmar@texsonics.net

www.texsonics.net

SPECIFICATIONS AND VISUALS MAY CHANGE WITHOUT NOTICE AS PART OF ONGOING PRODUCT IMPROVEMENTS. IMAGES ARE FOR REFERENCE ONLY.
UNAUTHORIZED REPRODUCTION OF THIS CATALOGUE IS PROHIBITED WITHOUT WRITTEN PERMISSION FROM TEXSONICS SYSTEMS INDIA PRIVATE LIMITED.

TSA700-6N

Texsonics Systems India Private Limited

1/6-1, Keerakaran Thottam, Keeranatham, Coimbatore 641035 • Ph: +91 94426 24304

TSA SERIES

4-AXIS SCARA • PAYLOAD 6 kg • REACH 700 mm

dharmar@texsonics.net • www.texsonics.net